Projector derivatives in DFT+eDMFT stress and forces

Bird's-eye equations, the meaning of the experimental NO_PROJECTOR test, and a cross-check against the established force derivation

Main conclusion

A genuine derivative of the DMFT projector cannot be supplied by the ordinary DFT functional. The force derivation confirms this: the analogous term is explicitly present and is printed as the orbital (`+ ORB`) force. Therefore the final stress formula must retain a projector-response term. The current discrepancy asks whether the implemented strain derivative is the correct fixed-coefficient derivative - not whether projector physics should be deleted.

This note separates exact stationary-functional identities from diagnostic observations. In particular, the excellent $SrVO_3$ result obtained by suppressing one implemented stress channel is treated as evidence that this channel is presently misdefined or mispartitioned, not as a derivation that the physical term vanishes.

Prepared from the LAPW force notes and the current `eDMFT_new/src/dmft2` implementation. No production source code was changed in preparing this note.

1. Definitions and conventions

At each k point, let ε_k be the diagonal Kohn-Sham Hamiltonian in the chosen band window, and let U_k be the band-to-correlated-orbital projector. The local self-energy entering the lattice problem is

$$\Sigma(i\omega) = \Sigma_{imp}(i\omega) - V_{DC}. \tag{1}$$

The embedded self-energy and lattice Green function are

$$\Sigma_k^{emb}(i\omega) = U_k\, \Sigma(i\omega)\, U_k^{\dagger}, \tag{2}$$

$$G_k^{-1}(i\omega) = (i\omega + \mu)\, I - \varepsilon_k - \Sigma_k^{emb}(i\omega). \tag{3}$$

The symbol λ denotes any external geometric parameter: an atomic displacement $R_{A\alpha}$ for a force, or a homogeneous strain component $\varepsilon_{\alpha\beta}$ for stress. All Matsubara sums below include temperature, k-point, spin, and degeneracy weights used by the code.

2. Stationarity and the one-particle projector gradient

The DFT+eDMFT Klein/Luttinger-Ward functional is stationary with respect to its internal variables at self-consistency. Consequently, when differentiating with respect to an external geometry, implicit variations of G, Σ, the density, and the impurity solution cancel against the corresponding functional derivatives. No two-particle vertex is required for the first derivative. The surviving projector dependence is explicit.

At fixed Σ, varying the embedded operator gives

$$\delta\Sigma_k^{emb} = (\delta U_k)\, \Sigma\, U_k^{\dagger} + U_k\, \Sigma\, (\delta U_k)^{\dagger}. \tag{4}$$

Hence the explicit projector contribution to the stationary functional is

$$\delta\Gamma_{proj} = 2\, \mathrm{Re}\, \Sigma_k\, \mathrm{Tr}_{loc,band}[\, C_k\, \delta U_k\,], \tag{5}$$

$$C_k = T \Sigma_\omega \Sigma(i\omega)\, U_k^\dagger\, G_k(i\omega). \tag{6}$$

The matrix C_k is the one-particle functional gradient conjugate to U_k. It is already frequency summed. In the current implementation it is reconstructed without a new Matsubara loop from the identity

$$U_k C_k = \rho_k^E - \varepsilon_k \rho_k = T \Sigma_\omega \Sigma_k^{emb}(i\omega) G_k(i\omega), \tag{7}$$

$$\rho_k = T \Sigma_\omega G_k,\ \rho_k^E = T \Sigma_\omega [\varepsilon_k + \Sigma_k^{emb}] G_k. \tag{8}$$

What stationarity cancels - and what it does not

Stationarity cancels the response $\delta\Sigma/\delta\lambda$ and the associated impurity vertex. It does not cancel the explicit geometrical change of the embedding $U_k(\lambda)$. Equation (5) is therefore a required one-particle term whenever the projector changes with geometry.

3. Normalized projectors

For projector types up to 5 in the present production path, the object used in DMFT is globally normalized:

$$U_k = P_k X,\ S = \Sigma_k w_k P_k^\dagger P_k,\ X = S^{-1/2}. \tag{9}$$

The k-point weights are fixed, so $w_{k,\lambda}=0$. The chain rule gives

$$\partial_\lambda U_k = (\partial_\lambda P_k) X + P_k (\partial_\lambda X). \tag{10}$$

The overlap derivative is

$$S_{,\lambda} = \Sigma_k w_k [P_{k,\lambda}^\dagger P_k + P_k^\dagger P_{k,\lambda}]. \tag{11}$$

For a matrix normalization, $X_{,\lambda}$ must be obtained from the Sylvester equation, not by differentiating diagonal entries independently:

$$X^{-1} X_{,\lambda} + X_{,\lambda} X^{-1} = -X\, S_{,\lambda} X. \tag{12}$$

This is the origin of the normalization term $P\, X_{,\lambda}$. Historically, the force implementation differentiated P and then multiplied by X, but did not include $X_{,A}$. The current code now includes this missing branch.

4. Fixed band-index window and the three projector responses

The DMFT window is a fixed set of band indices, not a fixed energy interval. Its membership does not change under an infinitesimal perturbation, but its eigenvectors do. It is useful to split the full derivative into three conceptually distinct pieces:

$$\partial_\lambda U = [P_{,\lambda}^{geom} X + P\, X_{,\lambda}^{geom}] + [P_{,\lambda}^{wf} X + P\, X_{,\lambda}^{wf}]. \tag{13}$$

- Explicit geometry: the LAPW basis, reciprocal vectors, matching coefficients, local projector, volume factor, and possibly local axes change at fixed eigenvector coefficients.
- Wavefunction/window response: the selected band subspace rotates into bands outside the window. This is the complementary-band or Sternheimer contribution.
- Normalization response: both geometry and wavefunction changes alter S and therefore $X=S^{-1/2}$.

For a window state n and an outside state a, the first-order eigenvector response has the familiar form

$$|\psi_{n,\lambda}\rangle_{outside} = \Sigma_{a \notin W} |\psi_a\rangle \langle\psi_a| H_{,\lambda} - \varepsilon_n O_{,\lambda} |\psi_n\rangle / (\varepsilon_n - \varepsilon_a). \tag{14}$$

The present implementation represents this response as auxiliary density and energy-density matrices, allowing the existing LAPW contractions to evaluate

$$\mathrm{Tr}[\rho^{\mathrm{resp}} H_{,\lambda}] - \mathrm{Tr}[\rho^{E,\mathrm{resp}} O_{,\lambda}]. \tag{15}$$

5. The unambiguous operator-level formula

A clean way to avoid bookkeeping ambiguity is to return to the nonorthogonal LAPW basis $\{\chi_K\}$. Let A contain the Kohn-Sham eigenvectors, $A^\dagger O A = I$, and define the embedded DMFT operator in this basis as

$$V^{\Sigma}_{KK'}(i\omega) = \langle \chi_K | \varphi_m \rangle \, \Sigma_{mm'}(i\omega) \, \langle \varphi_{m'} | \chi_{K'} \rangle. \tag{16}$$

At fixed Σ, the stationary derivative has the schematic structure

$$\delta\Gamma = \mathrm{Tr}[\rho^{\chi} \delta H^{0}] - \mathrm{Tr}[\rho^{E,\chi} \delta O] + T\Sigma_{\omega} \mathrm{Tr}[G^{\chi} \delta V^{\Sigma}] + \delta\Gamma_{\mathrm{DFT,ion}}. \tag{17}$$

Here δH^0 and δO are the usual LAPW Hamiltonian and overlap variations; $\delta\Gamma_{\mathrm{DFT,ion}}$ collects the Hartree-XC cancellation, nuclear term, and the standard surface and density-gradient terms. The third term is the explicit derivative of the embedded DMFT operator at fixed eigenvector coefficients. In band-projector notation it is Eq. (5).

Critical distinction

The overlap term in Eq. (17) contains a DMFT energy-density matrix and therefore has a ΣG component. Nevertheless, this is not the same object as δV^{Σ}. The former is the response of the nonorthogonal LAPW metric; the latter is the response of the correlated embedding operator. A true projector derivative cannot be declared to be 'already in DFT'.

6. Where a covariant-derivative question can enter

Writing the band projector schematically as $U = A^\dagger Q$ gives

$$\delta U = (\delta A^\dagger) Q + A^\dagger \delta Q. \tag{18}$$

The first term is eigenvector response. The second is the fixed-coefficient derivative of the basis-to-local projector matrix. Differentiating $A^\dagger O A = I$ shows that δA contains a metric connection proportional to $A^\dagger \delta O A$. If one constructs a quantity called δU that already contains this δA contribution and also adds the full $-\mathrm{Tr}[\rho^E \delta O]$ term, double counting is possible. This motivates a covariant-bookkeeping diagnostic.

However, the current `projector_strain` array is built with the LAPW eigenvector coefficients held fixed. At face value it therefore aims to represent $A^\dagger \delta Q$, not $(\delta A^\dagger) Q$. The force derivation likewise uses the full fixed-coefficient derivative of the projector matrix element. Consequently, a covariant subtraction is a hypothesis to test, not a justified production change.

$$D_\lambda U \equiv U_{,\lambda}^{\mathrm{constructed}} - (\text{metric/eigenvector connection already represented elsewhere}). \tag{19}$$

Equation (19) is useful only after the connection is identified algebraically. It is not permission to drop Eq. (5).

7. What the experimental NO_PROJECTOR mode actually omitted

The diagnostic switch set only the explicitly accumulated stress channel to zero. In mathematical terms it removed

$$\delta\Gamma_{\mathrm{explicit,implemented}} = 2\,\mathrm{Re}\,\Sigma_k \mathrm{Tr}[C_k (P_{k,\varepsilon}^{\mathrm{geom}} X + P_k X_{,\varepsilon}^{\mathrm{geom}})]. \tag{20}$$

It did not remove:

- the main $\mathrm{Tr}[\rho H_{,\varepsilon}] - \mathrm{Tr}[\rho^{E} O_{,\varepsilon}]$ LAPW stress;
- the complementary-band fixed-window response;
- the separate normalization-window response; or
- the fact that ρ and ρ^{E} themselves are DMFT quantities.

Interpretation

NO_PROJECTOR is a scalar/tensor isolation test. Agreement in one material means the implemented channel is compensating incorrectly there. It does not prove the exact physical channel is zero. The FeSe behavior already shows that simply suppressing this channel is not a universal solution.

8. Numerical diagnostics obtained so far

Test	Result	Meaning
$SrVO_3$ frozen-Σ numerical	mean 4.61820 GPa	Stable between $\|\varepsilon\|=0.001$ and 0.0005
$SrVO_3$ analytic, current full	mean 8.17689 GPa	+3.55869 GPa mismatch
$SrVO_3$ NO_PROJECTOR	mean 4.58436 GPa	Only -0.03384 GPa from numerical
Direct endpoint δU contraction	trace ≈ 4.9-5.1 GPa	Total projector response, gauge aligned
Implemented explicit channel	|trace| ≈ 10.78 GPa	Too large relative to inferred residual
FeSe RELA, old controlled set	numeric -1.0491; analytic -2.6416 GPa	Deleting explicit term would worsen old result

An internal finite difference of the code's own raw fixed-coefficient strain perturbation agrees with its analytic array to roughly 5×10^{-10} relative error. Therefore the algebra implemented for that internal perturbation is self-consistent. The remaining possibilities include an incorrect physical definition of the internal strain perturbation, an incorrect partition against window/normalization response, or endpoint gauge/completeness issues.

9. Force derivation: the projector term is present

The force notes derive the stationary DFT+eDMFT force in a nonorthogonal LAPW basis. Suppressing the ordinary DFT cancellations and writing only the structurally relevant terms, Eq. (326) of those notes is

$$\delta\Gamma = \mathrm{Tr}[\rho^{\mathrm{DMFT}} A^{\dagger}\delta H^{0} A] - \mathrm{Tr}[(\rho\varepsilon)^{\mathrm{DMFT}} A^{\dagger}\delta O A] + T\Sigma_{\omega}\mathrm{Tr}[G\, A^{\dagger}\delta V^{\Sigma} A] + \dots . \tag{21}$$

The third term is exactly the force analog of Eq. (5). It did not vanish. In the final force expression it appears as Eq. (364) of the notes, with the force sign $F_A = -\partial\Gamma/\partial R_A$.

$$F_A^{\mathrm{proj}} = -2\,\mathrm{Re}\,\Sigma_k \mathrm{Tr}[C_k\, \partial_{R_A} U_k]. \tag{22}$$

9.1 Why the displacement derivative becomes simple

For an atom-centered projector moving rigidly with its muffin-tin sphere, define the basis-space projector matrix

$$M_{KK'}^{A} = \langle \chi_K | \varphi_m^{A} \rangle \langle \varphi_{m'}^{A} | \chi_{K'} \rangle. \tag{23}$$

Differentiation contains derivatives of both LAPW basis functions, derivatives of the moving local orbitals, and a surface term because the integration domain moves. Combining them gives the rigid-translation identity derived as Eqs. (352)-(354) in the force notes:

$$\partial M_{KK'}^{A}/\partial R_{A\alpha} = i(K'_{\alpha} - K_{\alpha})\, M_{KK'}^{A}. \tag{24}$$

This result is important for the earlier question about a projector that is finite at R_{MT}. The boundary value does not by itself make the derivative disappear. The moving-domain surface term must be retained; in the rigid-translation calculation it combines with the volume derivatives to yield Eq. (24).

> **Force-specific simplification**
>
> Equation (24) follows from rigid translation. A homogeneous strain changes reciprocal vectors, angular directions, matching coefficients, the cell normalization, and possibly local axes. There is no general replacement $\partial_{\varepsilon} M = i(K-K')M$. Thus the force implementation validates the existence and contraction of the projector term, but it does not automatically validate the strain-specific formula for $P_{,\varepsilon}$.

9.2 How the force code represents it

The projector builder stores four channels when forces are active: U and three displacement channels $D_{A\alpha}$, with the convention

$$\partial_{R_{A\alpha}} U = -iD_{A\alpha}. \tag{25}$$

The derivative channels are formed by multiplying the LAPW expansion coefficients by the appropriate Cartesian K component. The compressed transformation then constructs the derivative of $U^{\dagger}GU$. Finally the Matsubara contraction with Σ produces `forb`, which is printed as `+ ORB`. In compact notation, this is Eq. (22).

This term was already present in the older implementation. The recent corrections did not invent it; they completed two pieces omitted by the historical construction:

- global normalization: $P\, X_{,A}$ was added to the old $P_{,A} X$ channel;
- finite band window: the complementary-band eigenvector response was added using the stored LAPW1 bands.

10. Side-by-side force and stress formulas

Object	Atomic force	Homogeneous stress
External variable	$R_{A\alpha}$	$\varepsilon_{\alpha\beta}$
Sign	$F=-\partial\Gamma/\partial R$	$\sigma=+(1/\Omega)\partial\Gamma/\partial\varepsilon$ in the WIEN2k comparison convention
Main LAPW part	$-\mathrm{Tr}[\rho H_{,A}]+\mathrm{Tr}[\rho^{E}O_{,A}]+\ldots$	$(1/\Omega)\{\mathrm{Tr}[\rho H_{,\varepsilon}]-\mathrm{Tr}[\rho^{E}O_{,\varepsilon}]+\ldots\}$
Projector part	$-2\,\mathrm{Re}\,C{:}U_{,A}$	$+(2/\Omega)\,\mathrm{Re}\,C{:}U_{,\varepsilon}$
Raw explicit derivative	Rigid-translation phase after surface cancellation	Reciprocal, angular, radial, volume, boundary, axes
Normalization	$P\,X_{,A}$	$P\,X_{,\varepsilon}$
Outside-window	Stored-band Sternheimer sum	Same response construction

11. What the force comparison teaches us about the stress bug

- The physical projector term must remain. Force theory and code both include its analog.
- The functional gradient C and the contraction $2\,\mathrm{Re}\,C{:}\delta U$ are shared by force and stress. A successful force test gives confidence in this half of the stress calculation.
- The raw geometric derivative is not shared: translation obeys Eq. (24), whereas strain requires a full metric deformation. This is now the most suspect part.
- Normalization and fixed-window responses are shared structurally and should be audited with the same finite-difference methodology in force and stress.
- A zero total force is a powerful acoustic sum-rule test. It does not prove that every strain term is correct, because uniform translation leaves the cell metric unchanged.

12. Recommended next derivation and diagnostics

12.1 Derive the strain term at the operator level

Start from Eq. (17), and differentiate the basis-space matrix $V^{\Sigma}_{KK'}$ directly under a homogeneous strain while holding the LAPW eigenvector coefficients fixed. Keep separately:

- the derivative of reciprocal vectors and spherical harmonics;
- the derivative of LAPW/APW+lo matching coefficients;
- the $1/\sqrt{\Omega}$ cell-normalization derivative;
- the derivative of the projector radial function and the muffin-tin boundary term at fixed R_{MT};
- the derivative of local correlated axes, initially zero for fixed Cartesian axes;
- the derivative of the global normalization X.

Only after this operator derivative is obtained should it be rewritten as $2\,\mathrm{Re}\,C{:}U_{,\varepsilon}$ and compared to the current `projector_strain` array.

12.2 Perform a scalar closure test

For each strain component q, compare three independently assembled quantities:

$$D_{total}^{FD}(q) = 2\,\mathrm{Re}\,C : [U(+h)-U(-h)]/(2h), \tag{26}$$

$$D_{wf}^{analytic}(q) = D_{window}(q) + D_{normalization\text{-}window}(q), \tag{27}$$

$$D_{explicit}^{expected}(q) = D_{total}^{FD}(q) - D_{wf}^{analytic}(q). \tag{28}$$

Then test directly

$$D_{explicit}^{expected}(q) \;?=\; 2\,\mathrm{Re}\,C : U_{,q}^{explicit,code}. \tag{29}$$

This comparison avoids the complete LAPW stress assembly and isolates precisely the disputed projector derivative. It should be repeated with increasing numbers of stored bands and with gauge alignment inside degenerate band blocks.

12.3 Use the force implementation as a control experiment

At a converged test structure, perform the analogous projector-only finite difference for an atomic displacement:

$$2\,\mathrm{Re}\,C : [U(R_A+h)-U(R_A-h)]/(2h) \;?=\; 2\,\mathrm{Re}\,C : U_{,A}^{force\ channels}. \tag{30}$$

This checks the derivative channels, normalization derivative, band-window response, signs, and units with a geometry for which Eq. (24) is known analytically. If this passes while Eq. (29) fails, the remaining error is unambiguously strain-specific.

13. Precise answer to the original question

Did the projector derivative vanish in the force calculation?

No. The force derivation contains it explicitly, and the implementation contracts it with ΣG to produce the orbital force. What simplified was the raw displacement derivative: after the moving-sphere boundary terms are included, it becomes the rigid-translation phase of Eq. (24). The newer force correction also adds the derivative of the nontrivial global normalization and the finite-window complementary-band response.

What should be changed in the stress formula now?

Nothing should be deleted on the basis of the $SrVO_3$ NO_PROJECTOR result alone. First derive and verify the correct fixed-coefficient strain derivative of the embedded operator. If the current array contains an extra metric/eigenvector connection, subtract exactly that identified piece; if it omits boundary, radial, or axis terms, add them. The final formula must still contain Eq. (5).

14. Mapping to the present source tree

Role	Current location
Build force and strain projector channels	src/dmft2/l2mainSubs.f90: Build_DMFT_Projector
Compress force derivative channels	src/dmft2/l2mainSubs.f90: CompressSigmaTransformation2
Contract orbital force	src/dmft2/cmp_log_gdloc.f90: forb / + ORB
Add P $X_{,A}$	src/dmft2/l2mainSubs.f90: AddProjectorNormalizationDerivative
Build window response and C	src/dmft2/l2mainSubs.f90: BuildWindowResponseMatrices
Contract explicit stress	src/dmft2/l2main.f90: stress_projector accumulation near line 923
Contract force window response	src/dmft2/cmp_mt_density.f90: cmp_MT_force_only

15. References

1. K. Haule, Some notes on forces in LAPW (9 November 2015), especially Sec. III and Eqs. (313)-(326), (352)-(365). Source: /Users/haule/projects/eDMFT/src/doc/Forces_LAPW.pdf.

2. Current working implementation under /Users/haule/projects/eDMFT_new/src/dmft2. The source tree contains ongoing diagnostic work; the equations in this note distinguish established identities from experimental implementation choices.

15.1 Status ledger

Established	Still diagnostic / unresolved
Stationarity removes the self-energy vertex from the first geometric derivative.	Whether the present strain array is the correct fixed-coefficient derivative of the embedded operator.
The explicit projector derivative survives and is present in the force derivation.	Whether any precisely identifiable metric connection is counted in two channels.
Force code contracts the displacement channels into the orbital (+ ORB) force.	The material dependence of the $SrVO_3$, FeSe, MnTe, and MnO discrepancies after fully controlled basis convergence.
Normalized projectors require both $P_{,\lambda}X$ and $P\,X_{,\lambda}$.	The complete strain-boundary and projector-5 radial contribution beyond the internal code perturbation check.

Decision gate: do not alter the production stress formula until the operator-level strain derivative and the scalar closure test, Eqs. (26)-(29), identify a specific missing or duplicated piece.